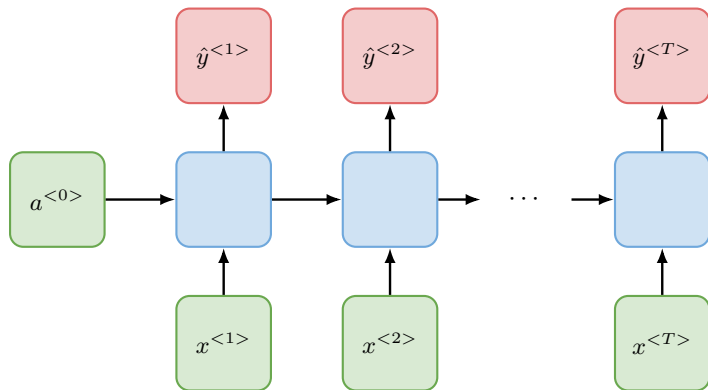


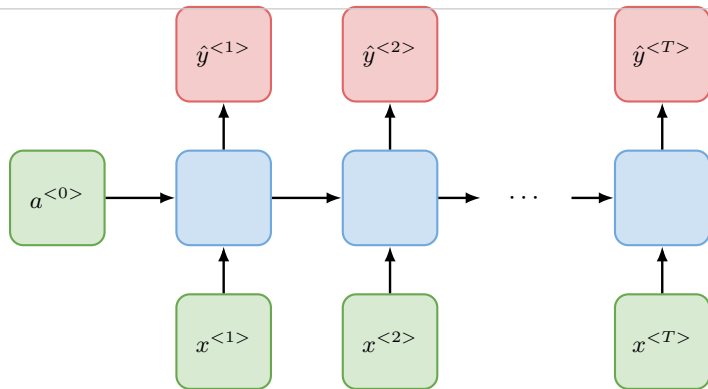
Many-to-many
 $T_x = T_y$
 one step = one trial
 T = trials in a session



Choice, and its timing, per trial

$x^{<t>} = (a_{t-1}, s_{t-1}, r_{t-1}, s_t, \text{cue}_t)$
 $\hat{y}^{<t>} = P(a_t)$ over Short / Mid / Long, plus decision
 time, execution time and takeoff heading
 (Section 8)
 $a^{<t>} = h_t$, $d = 1-4$ units
 every step has a label (a_t), so every $\hat{y}^{<t>}$ enters the loss

Many-to-many
 $T_x = T_y$
 one step = 10 ms
 $T \approx 335$ steps in a trial



Cursor trajectory within the trial

$x^{<t>} = (\mathbf{b}, \text{dir}, \text{go}(t))$; $\mathbf{b}$ is row 1's own trial-level read-out, held constant across these steps
 $\hat{y}^{<t>} = (x, y)(t)$, cursor position at step t
 $a^{<t>} = g(t)$, GRU with M units
 a measured position at every step: every $\hat{y}^{<t>}$ enters the loss

Notation: $x^{<t>}$ input, $a^{<t>}$ hidden state (h_t or $g(t)$ in the design document), $\hat{y}^{<t>}$ output, $a^{<0>}$ the learned initial state. $T_x = T_y$ in both: one output per input step.